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MIDTERM TEST

ECE 657A - Section 2 - Data and Knowledge Modelling and Analysis

University of Waterloo

Term: Winter Year: 2022

Date: Tuesday, March 1, 2022

Time: 10:00 – 11:20 pm (80 minutes)

Instructors: Prof. Mark Crowley

Lecture Section: 002

Exam Type: In person written test

Number of exam pages (includes cover page): 16

Additional Materials Allowed: Notes (3 pages, handwritten, double-sided) and calculator allowed

Please Print

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Section _____

Instructions

1. Answer the questions in the spaces provided. If you require additional space to answer a question, the last two pages and refer to those pages in your solutions. Do not tear off the pages in the booklet.
2. Use a dark pencil or pen for your work.
3. For multiple choice and true-false questions (A=True, B=False) fill in the right answer on the bubble sheet on the final page.
4. During the text, questions will only be answered for clarification of meaning or if you think there is a mistake. For anything more, or if the question is still unclear then state your assumptions and understanding of the question on the page and answer as best you can.
5. Washroom breaks: this is a short test, so hopefully it will not be necessary to be excused for a washroom break. However, if you need to, raise your hand and a TA will escort you to the outside of the washroom, leave your phone in your bag.
6. Additional Materials Allowed: Notes (3 pages, handwritten, double-sided) and calculator for page numbers.

True or False

Answer the following questions with either True or False. Enter your answer on the **bubble sheet** at the back of your exam, use "A" for True and use "B" for False. Note: you can circle the correct answer on the question sheet as well for reference, but be sure to fill in the correct option on the bubble sheet.

1. **(False)**: If your dataset is **large**, you would be better off with a model that has **low variance** and **high bias** rather than one with high variance and low bias.
2. **(True)**: If your dataset is **small**, you would be better off with a model that has **low variance** and **high bias** rather than one with high variance and low bias.
3. **(False)**: Computing the MLE for linear regression can yield multiple local optima.
4. **(True)**: A MAP estimator is equivalent to the MLE estimator when the prior used in the MAP is a uniform prior over the parameter space.
5. **(False)**: AdaBoost will continue improving indefinitely eventually give zero training error regardless of the type of weak classifier it uses, provided enough iterations are performed. **Answer:** It will continue to improve, but it might not hit zero error if the weak classifier is limited in a particular way to the data, see EXOR example.

1: This may not be easy to see, and it depends on what 'best' means. But large data won't fix bias anyways, while it can reduce your variance if you have enough data. So you could argue it's better to predict something tightly, and then correct for the bias later.

Multiple Choice

Choose the correct answer to the following questions. Enter your answer on the **bubble sheet** at the back of your exam for the corresponding question number and letter. Note: you can circle the correct answer on the question sheet as well for reference, but be sure to fill in the correct option on the bubble sheet.

6. Which is of the following situations gives you better confidence your model is learning well?
- (A) A classifier that achieves 94% accuracy on the training set and 70% accuracy on test set
 - (B) **Answer: (correct)** A classifier that achieves 70% accuracy on the training set and 75% accuracy on test set.
7. What concept does the previous demonstrate about your model? (write the answer on this sheet)
- Answer: generalizability**

Because it could have learned a general pattern that explains the generating distribution well, doing better on test data than on it's training data. (B) would be a rare situation, but (A) is doing so much worse on the test set than on the training set that it is likely overfitting.

8. When data is divided into training, validation and testing sets, and changing some parameter is leading to improved accuracy on the training set, but decreasing accuracy on the validation set, then what is the explanation?
- (A) there is not enough data for training
 - (B) the model is biased in some way
 - (C) the data probably has too many outliers
 - (D) **Answer: (correct)** the model is overfitting to the training data
 - (E) the model is underfitting to the training data
9. MAP estimators are less likely to suffer from over-fitting compared to MLE estimators due to which of the following?
- (A) the use of Bayes rule
 - Answer: (correct)** the prior distribution
 - (B) the use of the expectation operator
 - (C) no assumption about features being independent
 - (D) the posterior distribution

A number of people attempted to answer 8,9,10 or 11 with more than one answer. Unless explicitly stated, multiple choice questions have only one correct answer. Guessing multiple answers results in a zero.

Multiple Choice Data Preparation

We learned about a number of data pre-processing methods in class, such as the following:

- (A) Cross-validation
- (B) Histograms
- (C) Normalization
- (D) Interpolation
- (E) One-hot Encoding

For each of the numbered tasks or situations below, choose the letter corresponding to the most appropriate data pre-processing method to use:

- 10. Using both Nominal and Continuous Data

(A)	(B)	(C)	(D)	(E)
-----	-----	-----	-----	-----

Answer: (E)
- 11. Noisy Data

(A)	(B)	(C)	(D)	(E)
-----	-----	-----	-----	-----

Answer: (B)
- 12. Missing Data

(A)	(B)	(C)	(D)	(E)
-----	-----	-----	-----	-----

Answer: (D)
- 13. Finding Outliers

(A)	(B)	(C)	(D)	(E)
-----	-----	-----	-----	-----

Answer: (B)
- 14. Different Data Scales

(A)	(B)	(C)	(D)	(E)
-----	-----	-----	-----	-----

Answer: (C)
- 15. Reducing Bias of Your Model

(A)	(B)	(C)	(D)	(E)
-----	-----	-----	-----	-----

Answer: (A)

Multiple Choice Algorithm Assessment

You are asked to prepare an assessment of the available methods to use for the following problems. For each numbered problem, there are a number of choices indicated by letter in the table. Check off the appropriate selection for each. Then go to the bubble sheet at the end of the test booklet and fill in the appropriate lettered boxes for each numbered question.

The table options are as follows, select one case only for each algorithm. Provide a one line explanation for cases (B) and (C) only:

- (A) This method could be used, without additional assumptions.
- (B) This method could be used, with some additional assumptions (explain).
- (C) Not applicable to this task for specific reasons (explain).
- (D) This method doesn't do the desired task at all

II: Sensor data on energy, heat, noise and speed for a factory assembly line motor, where the general relationship between the variables is known. The task is to monitor the outcoming sensor data and learn a model that can predict one of the variables from the other three. **Answer:** This question was a bit confusing for people, so we'll great it manually and consider the quality of explanations.

16. MLE

(A)	Answer: !	(B)	(C)	(D)
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17. Explanation (if needed): **Answer:**

18. MAP

(A)	(B)	Answer: !	(C)	(D)
-----	-----	------------------	-----	-----

19. Explanation (if needed): **Answer:** prior for parameters needs to be specified

20. EM

(A)	(B)	(C)	(D)
-----	-----	-----	-----

21. Explanation (if needed): **Answer:** B or C is correct, given appropriate reasoning. A or D should be marked as incorrect

22. Naive Bayes

(A)	(B)	Answer: !	(C)	(D)
-----	-----	------------------	-----	-----

23. Explanation (if needed): **Answer:** any mention of the parameters not being independent, prior for parameters needs to be specified, extra point if answer is same as MAP

24. PCA

(A)	(B)	(C)	(D)	Answer: ?
-----	-----	-----	-----	------------------

25. Explanation (if needed): **Answer:** C can also be correct if mentioned the correct reason

26. LDA

(A)	(B)	(C)	(D)	Answer: ?
-----	-----	-----	-----	------------------

27. Explanation (if needed): **Answer:** Can also be B, if discretization/classification assumption is mentioned

28. kNN

(A)	(B)	(C)	(D)
-----	-----	-----	-----

Answer: !

29. Explanation (if needed): **Answer:** Can also be B, if discretization/classification assumption is mentioned

30. t-SNE

(A)	(B)	(C)	(D)
-----	-----	-----	-----

Answer: !

31. Explanation (if needed): **Answer:**

Pre-processing with PCA

Consider the use of PCA as a pre-processing step on your data for a classification task and answer the following questions briefly. Let our dataset be a moderate size dataset $\mathbf{X} \in \mathbb{R}^d$ with 50 features, which are very human interpretable, a numerical label that relates in a complex, non-linear way to the features, and 10,000 data points. We apply PCA to the data obtain the eigenvector matrix $\mathbf{U} \in \mathbb{R}^{d \times d}$ without truncation.

We then use the projection $\mathbf{U}^T \mathbf{X}$ to train each of the following models for our classification problem, we compare the accuracy of the classifier against being trained on $\mathbf{X}$ directly.

32. Naive Bayes: What is one benefit and one drawback of using this PCA projection in this case?

Since Naive Bayes assumes that features are independent, after applying PCA to the data, this is more likely to be true since the eigenvectors are optimized towards being orthogonal (even though this isn't guaranteed unless the data is Gaussian). A drawback will be interpretability will be lost, NB has a straightforward probabilistic relationship with the features so it is quite interpretable usually.

33. Decision Tree using C4.5: What is one benefit and one drawback of using this PCA projection in this case?

The features being used by the decision tree will no longer have any semantic connection to the data, so interpretability is lost. A benefit could be that since the principle components are more representative of the variation in the data, fewer features and fewer splits will be needed to achieve the same result, so the model may overfit less. However, it should be noted that decision trees don't require features to be transformed in this way in order to find the best model.

34. Random Forests with 100 tree: What is one benefit and one drawback of using this PCA projection in this case?

Answer: The features being used by the decision tree will no longer have any semantic connection to the data, so interpretability is lost. In Random forests and kNN this is already very weakened so there is no loss from using it. There will be additional computation cost but it probably is worth it. Since the dimensionality is the same, we are not gaining performance from a faster model. Random forests will not benefit greatly from this, since they aren't carve out their own distinctions in features space, but kNN and Naive Bayes will benefit greatly.

Confusion Matrix Calculations

Calculate the answer and enter in the answer box with at least two decimal points of precision.

Confusion Matrix

Imagine you have built a classifier for identifying cars, and it correctly identifies 41 cars and 56 trucks. However, it incorrectly labels 8 cars as trucks, and it incorrectly labels 12 trucks as cars.

35. Draw the confusion matrix for this situation where the "positive" class is cars.

The tp of the classifier is 41.
The tn of the classifier is 56.
The fn of the classifier is 8.
The fp of the classifier is 12.

Confusion Matrix		
1=cars; 0=trucks		
	True Value	
	1	0
Estimated	-----	
1	41	12
0	8	56

Accuracy

36. What is the accuracy of the classifier? Show your calculation and the answer. **Answer:** The accuracy of the classifier is 0.83.

Also show formula and calculate it.

Recall

37. What is the the recall for the classifier? Show your calculation and the answer. **Answer:** The recall of the classifier is 0.84.

Also show formula and calculate it.

Representation Learning

38. Why is Singular Value Decomposition (SVD) an important alternative way to perform Principle Component Analysis (PCA)?
- (A) SVD is an approximation of the PCA solution, so it is more generalizable.
 - (B) SVD doesn't require the data matrix to be centred.
 - (C) SVD can deal with missing data.
 - (D) **Answer: (correct)** SVD is more computationally efficient to compute than PCA in some circumstances.

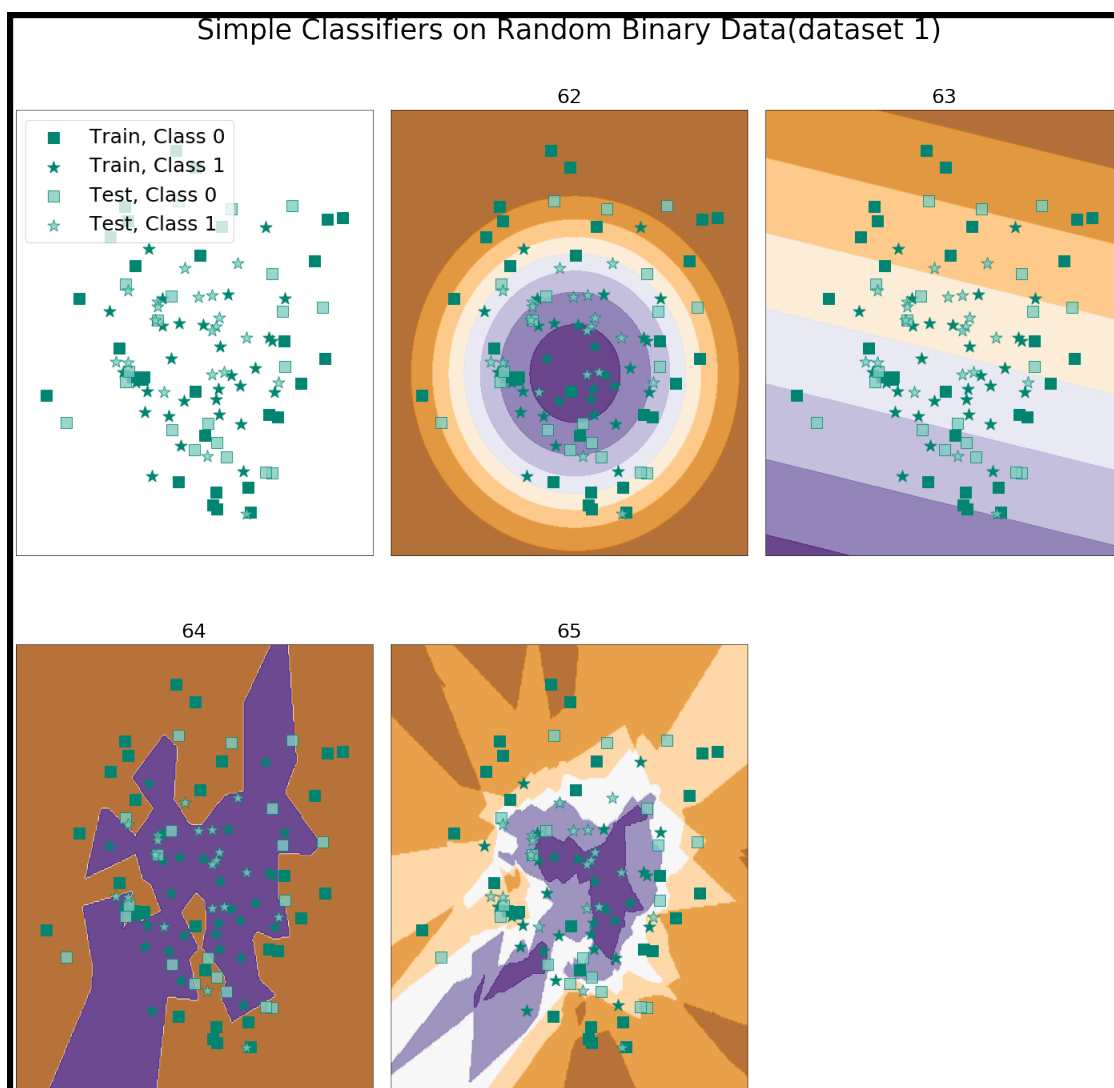
FDA vs. LDA

39. What is the difference between FDA and LDA at a high level?
- (A) FDA is supervised and LDA is not.
 - (B) **Answer: (correct)** LDA uses the features extracted by FDA to define a classifier.
 - (C) LDA focusses on classification whereas FDA focusses on separation and variance.
 - (D) LDA is computationally more efficient than FDA.

39: Most people said C, why?
Is it true in a sense?

Algorithm Decision Boundaries

The first plot is the original dataset. The next four plots numbered as Questions¹ were generated by one of the algorithm configurations below. Fill in the appropriate box in the multiple choice bubble sheet at the end of the test booklet. (A) Naive Bayes (B) LDA (C) kNN: k=1 (D) kNN: k=5 (E) Decision Tree: maxDepth=6

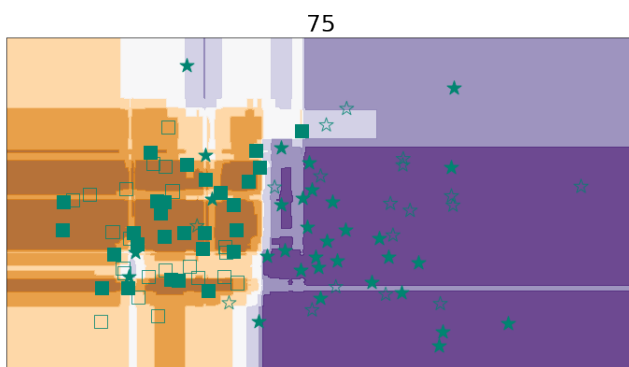
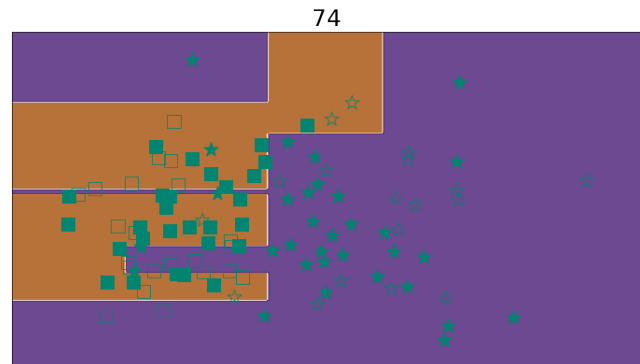
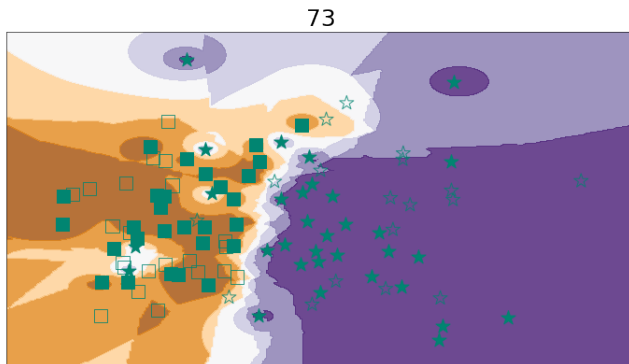
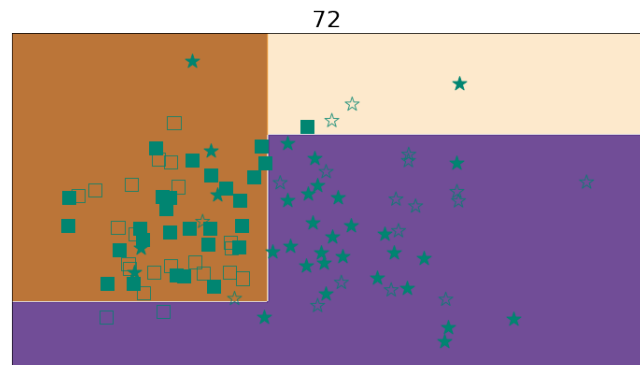
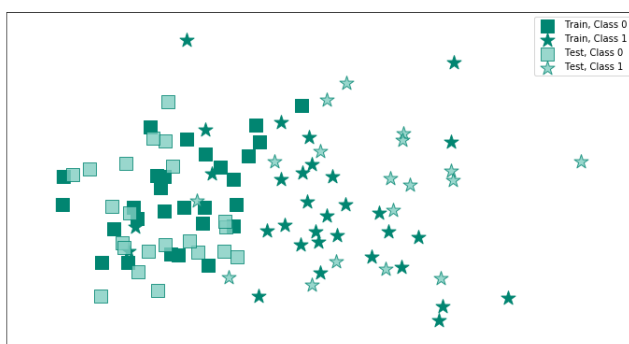


Answer: 62: (A) 63: (B) 64: (C) 65: (D)

¹Don't worry that the question numbers are higher, there's still room on the bubble sheet. Empty question numbers won't be used.

The first plot is the original dataset. The next four plots numbered as Questions² were generated by one of the algorithm configurations below. Fill in the appropriate box in the multiple choice bubble sheet at the end of the test booklet. (A) RF: numTrees=100, maxDepth=6 (B) Decision Tree: max depth=2 (C) Decision Tree: maxDepth=6 (D) kNN: k=5 dist=Euc (E) LDA

Complex Classifiers on Random Binary Data(dataset 2)



Answers:

72: B
73: D
74: C
75: A

²Don't worry that the question numbers are higher, there's still room on the bubble sheet. Empty question numbers won't be used.

Extra pages for answers or rough work. Please specify the question number here and the use of this page on the question page.

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